



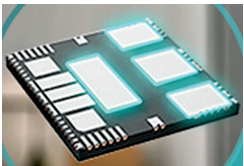
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Industry's first intelligent power module

Texas Instruments (TI) has launched a 650V three-phase GaN IPM for 250W motor drive applications. The IPM offers over 99% inverter efficiency, optimised acoustic performance, reduced size, and lower costs. It meets



energy-efficiency standards, enhances thermal performance, reduces power losses by 50%, and achieves low dead time and low propagation delay, resulting in quieter operation. The DRV7308 also enables higher PWM switching frequencies, reducing audible noise and system vibration, while decreasing motor heating, enhancing reliability, and extending system lifetime. The power module delivers high power density in a 12mm-by-12mm package, making it the smallest IPM for 150W to 250W motor-drive applications. It eliminates the need for an external heatsink, reducing motor drive inverter PCB size by up to 55%. Integration of features like a current sense amplifier and protection features further reduces solution size and cost.

Texas Instruments (TI)
<https://www.ti.com>

World's first RISC-V multi-mode LTE chipset

GCT's GDM7243SL is the world's first highly integrated multi-mode LTE chipset capable of operating from 450MHz down to 380MHz. Equipped with dual receive paths and low-band dual transmitters, it utilises 2 × 2 MIMO diversity and increased power to ensure

uninterrupted connectivity even in challenging environments. Its RISC-V dual-core design, featuring integrated



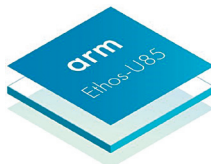
RAM and optional Flash, empowers users to craft fully integrated devices without external processors. Smartphones, tablets, IoT

devices such as smart meters, asset trackers, and environmental sensors, as well as industrial machinery and equipment in sectors like agriculture, mining, and construction, can incorporate the chipset to maintain connectivity in remote or underground locations.

GCT Semiconductor
<https://www.gctsemi.com>

Highest performing NPUs for IoT markets

ARM has developed Ethos NPUs to enhance AI performance at the edge and endpoints. The NPUs offer performance and energy efficiency for AI applications.

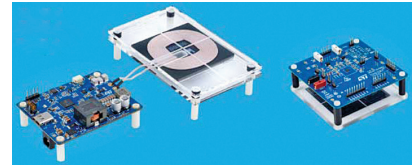


Ethos-U55 is integrated into Cortex-M based systems, while Ethos-U65 extends this to Cortex-A based systems, doubling on-device machine learning (ML) performance. The NPU family offers improved performance and energy efficiency. With support for transformer architecture networks and various AI operations, Ethos-U85 addresses the increasing demand for performance in IoT applications such as factory automation and smart home cameras. It also supports element-wise operator chaining, enhancing efficiency by reducing data transfer between the NPU and memory.

ARM
<https://www.arm.com>

High-power wireless charging solution

STMicroelectronics has launched a 50W transmitter and receiver pairing compatible with Qi standards, aimed at accelerating wireless charging in high-power applications like medical and industrial equipment, home appliances, and computer peripherals. This solution



enables the convenience and efficiency of wireless charging in sectors requiring higher power and quicker charging intervals. Applications such as vacuum cleaners, cordless power tools, drones, medical devices, portable ultrasound systems, stage and mobile lighting, printers, and scanners can benefit from this technology, eliminating the need for cables, connectors, and complex docking setups, resulting in simpler, more cost-effective, and robust products. The transmitter board can deliver up to 50W of output power, surpassing standard smartphone protocols for faster recharging of larger batteries. The receiver board incorporates adaptive rectifier configuration (ARC) to extend charging distance by up to 50%, enhancing coil flexibility and cost efficiency.

STMicroelectronics
<https://www.st.com/content>

Industry's lowest power 16-lane retimer

Marvell Technology, Inc. has introduced the Alaska P PCIe retimer product line to enhance data centre compute fabrics in various server types and infrastructures. These retimers maintain connection distances at high speeds, which